agrochemicals

MDPI

Article

Effects of Selected Biopesticides on Two Arthropod Pests of *Cannabis sativa* L. in Northeastern Oregon

Tiziana Oppedisano [1,†], Silvia I. Rondon [2,*] and Daniel I. Thompson [1]

1 Hermiston Agricultural Research and Extension Center, Oregon State University, Hermiston, OR 97838, USA; tiziana.oppedisano@montana.edu (T.O.)
2 Oregon Integrated Pest Management Center, Oregon State University, Corvallis, OR 97331, USA
* Correspondence: silvia.rondon@oregonstate.edu
† Current address: Department of Plant Sciences & Plant Pathology, Montana State University, Bozeman, MT 59717, USA.

Abstract

Hemp (*Cannabis sativa* L.) cultivation in the United States has expanded rapidly over the past decade. Due to federal and state regulations, only a limited number of studies have examined the chemical options available for controlling pests on *C. sativa*. In the U.S., two of the most important species of arthropod pests affecting *C. sativa* are the beet leafhopper *Circulifer tenellus* Baker (Hemiptera: Cicadellidae) and the two-spotted spider mite *Tetranychus urticae* Koch (Acari: Tetranychidae). This study evaluated the effects of four biopesticides, *Chromobacterium subtsugae*, *Burkholderia* spp., *Chenopodium ambrosioides*, and azadirachtin, under greenhouse conditions against *C. tenellus* adults and nymphs and *T. urticae* adults. Biopesticides were applied to foliage using a calibrated hand sprayer. To evaluate the biopesticides' potency, *C. tenellus* adults, nymphs, and mites were released 1 h after treatment; to evaluate the residual efficacy, they were released 7 days after treatment (DAT). In both experiments, *C. tenellus* adults, nymphs, and mites were counted 1, 3, and 7 days after release. Our results indicate that *Burkholderia* spp. exhibited the highest efficacy against *C. tenellus* adults at 7 DAT, whereas *C. ambrosioides* and azadirachtin caused the greatest nymphal mortality at 1 and 3 DAT, respectively. Our results show that *Burkholderia* spp. had the greatest potency against *C. tenellus* adults 7 DAT, while *C. ambrosioides* and azadirachtin highly affect the mortality of nymphs at 1 and 3 DAT, respectively. Treatments with *C. subtsugae* and *C. ambrosioides* showed high potency against *T. urticae*. Finally, *C. subtsugae* showed the lowest residual effect against the mite pest. The data presented in this article will add to the arsenal of information to improve the current management strategies used against these two hemp pests.

Keywords: *Cannabis sativa*; beet leafhopper; two-spotted spider mites; biopesticides; hemp; IPM; control strategies

Academic Editors: Yiming Wang, Danyu Shen and Wei Yan

Received: 28 August 2025
Revised: 17 October 2025
Accepted: 21 October 2025
Published: 26 October 2025

Citation: Oppedisano, T.; Rondon, S.I.; Thompson, D.I. Effects of Selected Biopesticides on Two Arthropod Pests of *Cannabis sativa* L. in Northeastern Oregon. *Agrochemicals* **2025**, *4*, 19. https://doi.org/10.3390/agrochemicals4040019

1. Introduction

Hemp (*Cannabis sativa* L.) is one of the oldest species to have been cultivated around the world [1]. Byproducts of this crop are used mainly as a source of fiber but also for oil and other components [2]. In Europe, hemp was widely grown between the 16th and 18th centuries [3], and it is believed to have been first introduced to North America in 1606 [3]. The significance of hemp is rapidly growing in the healthcare sector on account of its medicinal properties, which are widely researched and continue to stir much controversy [4]. The Farm Act of 2014 authorized the U.S. to establish pilot research programs for the cultivation of industrial hemp, thereby increasing knowledge about hemp cultivation in the grain, fiber, and oil industries [5]. This offers the potential to incorporate hemp as a rotational crop alongside other field crops and vegetables, allowing farmers to diversify their production and consequently improve sustainability however, much work remains to be conducted to understand the agronomic benefits of incorporating hemp into existing crop rotations. Today, hemp is considered a valuable crop in the U.S., and it is quickly becoming well-established in the U.S. Pacific Northwest, especially in Oregon. The region harbors over two hundred crops in the lower and upper Columbia Basin of Oregon and Washington State, where the climate is suitable for growing hemp.

Several arthropod pests, including leafhoppers and mites, affect hemp plants grown both indoors and outdoors, causing damage to various parts of the plant [6]. The beet leafhopper, *Circulifer tenellus* Baker (Hemiptera: Cicadellidae), is a generalist species that feeds and reproduces on a wide range of plants, including sugar beet (*Beta vulgaris* L.), potato (*Solanum tuberosum* L.), tomatos (*S. lycopersicum* L.), cucurbits (*Cucurbita* spp.), and various wild hosts [7]. The peculiarity of this pest is its ability to transmit plant pathogens to several hosts, including the *Beet Curly Top Virus* (BCTV), a *Curtovirus* with a small genome belonging to the Geminiviridae family. Even though the transmission dynamics of BCTV in hemp are still unclear, *C. tenellus* is the only known vector of this pathogen [8]. The BCTV has been reported in hemp in several states, such as Arizona, Colorado, and Nevada [9–11]. In Oregon, BCTV in hemp was found sporadically around 2018, but it has become more common in recent years [12]. To date, there is no documentation available regarding the incidence and population dynamics of *C. tenellus* on hemp in the region. Data collected on potatoes suggests that this species shows upward and downward trends over the years [7]. In Northeast Oregon, *C. tenellus* emerges when the temperature reaches 582 degrees, and 90% of pest emergence happens on 2823-degree days [13]. Therefore, *C. tenellus* moves freely after spring emergence between overwintering hosts such as Russian thistle (*Kali tragus* L.) and kochia (*Bassia scoparia* L.) to crops like potatoes or hemp.

The two-spotted spider mite, *Tetranychus urticae* Koch (*Acari*: *Tetranychidae*), is extremely polyphagous, with a strong ability to move between crops and fields [14]. By far, *T. urticae* is the most important species in greenhouses and other indoor environments, nationally and worldwide [15]. *Tetranychus urticae* feeds on the mesophyll on the underside of leaves, causing a wilted appearance, especially on the upper 1/3 of the canopy of crops. According to Attia et al. [16], typical *T. urticae* damage is described as "stippling on the upper surface of leaves", since spider mites feed mainly on the underside. At high levels of infestation, the plant reacts by dropping affected leaves, thus reducing the growth of flowers and leaves [17]. Webs produced by *T. urticae* can cause additional damage to the plant, since the web serves as a shelter from predators, protecting them against abiotic factors, and also serves as a vehicle for dispersion [14]. In addition, *T. urticae* is well known for its rapid development of resistance to pesticides [18]. Thus, new chemicals that can be used in rotational chemical programs are needed.

Generally, pesticides play a crucial role in pest control programs. The basic biological activity profile of pesticides begins with well-researched laboratory bioassays [19]. In the U.S., pesticides are regulated under the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA), as outlined in Title 40 of the Code of Federal Regulations. Each state regulates the use of synthetic pesticides for hemp production, including for medical and recreational purposes [20]. However, since December 2019, the U.S. Environmental Protection Agency (EPA) has approved the addition of 59 pesticide products that can be used on hemp, 58 of which are biopesticides and one of which is potassium salts of fatty acids, a conventional pesticide [21]. Biopesticides are pest control products that are generally considered to have reduced risks. The U.S. EPA defines biopesticides as pesticides derived from natural materials, divided into three categories: biochemical, microbial, and plant-incorporated protectant pesticides [22]. As hemp is a relatively recent legal crop in the U.S., State Departments of Agriculture, including Oregon's, conduct monthly reviews of pesticide labels to determine their eligibility for the hemp guide list within their pesticide programs.

Due to the presence of *C. tenellus* and *T. urticae* in various crops around Oregon, this study was conducted to determine the efficacy and residual activity of four biopesticides against these two pests of hemp, specifically in eastern Oregon. Our goal is to enhance current management practices by incorporating additional tools, aiming to target arthropod pests in a timely manner and minimize the severity of their damage, thereby maximizing hemp production.

2. Materials and Methods

2.1. Plant Material

Hemp seeds (var. White CBG) were obtained from a local hemp seed company (Oregon CBD Seeds, Independence, OR, USA). Each seed was planted in small pots (75 cm^3) using potting mix as a substrate (Kopacz Nursery, Echo, OR, USA) and Miracle-Gro fertilizer (The Scotts Miracle-Gro Company, Marysville, OH, USA). A 12:1 potting soil and sand mixture was used for all planting needs. Plants were irrigated three times per week using a watering can to moisten the top 2 cm of the soil. After approximately three weeks, when the plantlets were 15 cm tall, each plant was transplanted into 3.76 L pots and grown in the greenhouse at the Hermiston Agriculture Research and Extension Center (HAREC). The greenhouse was maintained at 25 ± 1 °C, with a photoperiod of 16:8 (L:D) hours and a humidity rate of 45–50%. No pesticides were applied to the plants before this experiment.

2.2. Circulifer tenellus

Circulifer tenellus specimens were initially obtained during the summer of 2020. Natural populations were collected on potatoes, Russian thistle, and kochia found around fields in Hermiston, OR, USA (45°49′15.1″ N, 119°17′00.0″ W), and Boardman, OR, USA (45°45′00.4″ N 119°57′45.9″ W). *Circulifer tenellus* colonies were maintained in a rearing room at the Irrigated Agricultural Entomology Program facility, HAREC, at 26 ± 2 °C and a photoperiod of 16:8 (L:D) hours. Insects were mostly reared on sugar beet, which is the primary host of this species; new plants were provided every other week to sustain large and healthy colonies. For this experiment, hemp plants were added to the rearing cages two weeks before the experiment began, allowing a transition to a new source of food. Males and females, and third- and fourth-instar nymphs, were used for the experiment.

2.3. Tetranychus urticae

A colony of *T. urticae* was maintained in a rearing facility at HAREC. Mites were collected from the field during the summer of 2020 in Hermiston, primarily on wild hosts, including several *Brassica* spp., and cultivated corn (*Zea mays* L.). The mite colony was maintained on 3.76 L potted bean plants (*Vicia faba* L.). As for *C. tenellus*, hemp plants were added to the rearing cages two weeks before starting the experiments. The environmental and growing conditions were the same as described above for *C. tenellus*. Only *T. urticae* adults were used during the experiments.

2.4. Biopesticide Treatments

Four biopesticides with different modes of action and an untreated control were evaluated and compared to assess their effects on *C. tenellus* and *T. urticae*. The four active ingredients of the biopesticides tested in this study included *Chromobacterium subtsugae* strain PRAA4-1T (30%), heat-killed *Burkholderia* spp. strain A396 cells (94.5%), *Chenopodium ambro-*

sioides (16.8% plant extract of *Chenopodium ambrosioides* near *ambrosioides*), and azadirachtin (1.2%). These biopesticides were selected based on their current registration status and increasing interest in their potential application to other economically important crops cultivated in the same region, such as potatoes. Table 1 summarizes information about the mode of action and rates of these biopesticides. Treatments were applied to the foliage when plants were 50 cm tall. The one-time applications were made using a calibrated 0.75 L hand sprayer. All biopesticide treatments were applied at concentrations corresponding to label recommendations, and solutions were prepared in distilled water on the day of the experiment. Insecticides were prepared separately in water containing 0.25% (*v*/*v*) surfactant.

Table 1. List of biopesticides used in this study against *Circulifer tenellus* and *Tetranychus urticae*.

Active Ingredient	Biopesticide Type	Mode of Action	Recommended Dose
Chromobacterium subtsugae strain PRAA4-1T (30%)	Microbial	repellency, ingestion, and reproduction disruption	4 g/L
Burkholderia spp. strain A396 (94.5%)	Microbial	exoskeleton degradation and molting interference	10 mL/L
Chenopodium ambrosioides (16.8%)	Biochemical	disruption of mobility and respiration	37.5 mL/L
Azadirachtin (1.2%)	Biochemical	repellency, anti-feedance, and molting interference	15 mL/L

2.5. Experiment Setup and Mortality Evaluation

This work was conducted in two consecutive experiments. Initially, we aimed to assess the potency effect of the four biopesticides. To achieve this goal, *C. tenellus* and *T. urticae* were released on hemp plants the same day of treatment, as soon as the foliage dried out after application. Following the Echegaray et al. [23] protocol, *C. tenellus* individuals were isolated in clip cages with small enclosures consisting of two small plastic cups (one inverted) attached to a clip to allow for close-fitting opening and closing. *Circulifer tenellus* adult and nymph trials were carried out in two different experiments. Five adults or five nymphs were released in each cage. Individuals of *C. tenellus* were collected from the colony using an insect aspirator and placed in a modified 200 mL pipette tip, which was inserted into a hole on one side of the clip cage. When nymphs were used, third- and fourth-instar individuals were collected individually from the colony and transferred onto the hemp plants. The youngest stages were discarded to avoid body damage and premature mortality, while fifth-instar insects were avoided to prevent development to adulthood during ongoing experiments. *Tetranychus urticae* adults were released in modified clip cages, which were slightly different and smaller than the ones used for *C. tenellus*. A detailed description and picture of the cages can be found in the Supplementary Material File S2 attached to this manuscript. A fine paintbrush was used to move the mites from the colonies to the cage. The mites placed on clip cages were set on hemp leaves by using a hand lens and a fine paintbrush. Spider mites were considered dead if no movement was observed after repeated gentle probing with a brush. To record the mortality rate, we counted dead *C. tenellus* and *T. urticae* individuals 1, 3, and 7 days after treatments (DAT). Clip cages containing dead and alive leafhoppers and mites were recollected and stored at −4 °C to prevent secondary infestation during the ongoing experiments.

In a second experiment, we aimed to assess the residual effects of the four biopesticides on the same previously treated plants. To achieve this goal, *C. tenellus* and *T. urticae* were released for a second time 7 DAT. For the release and the mortality evaluation, we followed the same method as that for the potency experiment. The mortality rates were evaluated by counting the number of dead individuals. Thus, *C. tenellus* adults and nymphs and *T. urticae* were counted 1, 3, and 7 days after rerelease, corresponding to 8, 10, and 14 DAT, respectively. Clip cages containing dead and alive leafhoppers and mites were recollected and stored at −4 °C to prevent a secondary infestation in the greenhouse.

In all experiments, four cages were attached to each plant (four plants per treatment), resulting in a total of 16 replicates per treatment. Moreover, to minimize positional effects and environmental variability, the treated plants were arranged in the greenhouse according to a randomized block design.

2.6. Data Analysis

Statistical analyses were performed using IBM SPSS Statistics v.20 (IBM Corp., Armonk, NY, USA). To determine the potency and residual effects, a one-way analysis of variance (ANOVA) was conducted. Levene's test of equality of error variances was implemented to test the assumption of equal variances of the ANOVA. Mortality rate data were transformed for normalization using the arcsine square root transformation procedure before analysis, calculated as $Y = arcsin(\sqrt{(x/100)})$, where x represents the percentage value. Tukey's comparison test was used to test for significant differences in mortality rates within the exact date of data collection (i.e., the time of treatment). Tukey's HSD test was performed at a significance level of $\alpha = 0.05$.

3. Results

The results of the potency and residual efficacy based on the cumulative percent mortality over time for *C. tenellus* adults, *C. tenellus* nymphs, and *T. urticae* are presented in Figure 1. Statistics, as presented in Figure 1, refer to untransformed data.

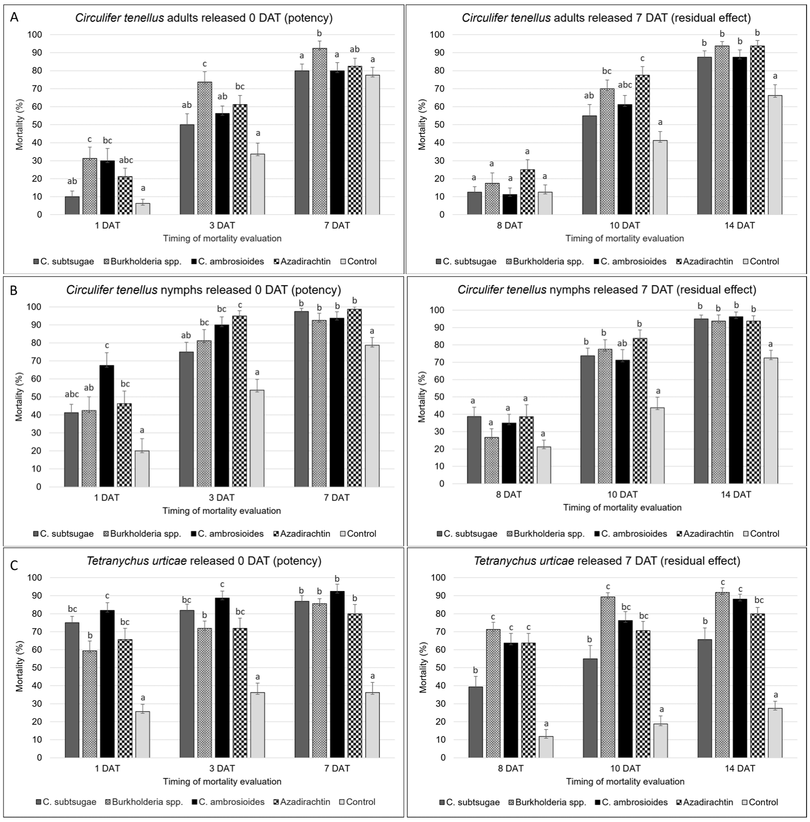

Figure 1. Effect of biopesticides on *Circulifer tenellus* adults (**A**) and nymphs (**B**) and *Tetranychus urticae* (**C**). The column on the left shows the potency effect results at 1, 3, and 7 DAT with arthropods released 0 DAT. The column on the right shows the residual effect results at 8, 10, and 14 DAT with arthropods released at 7 DAT. Data are presented as a percentage of mortality. The values represent the means ± standard error ($n = 16$). Columns within the same graph topped by different letters are significantly different from each other within the same date of data collection ($p < 0.05$). DAT: Days after treatment.

3.1. Effect of Biopesticides on *Circulifer tenellus* Adults

In the potency effect experiment, the mortality associated with the application of *Burkholderia* spp. at 1 DAT (31.2%) was significantly greater than the control and *C. subtsugae* ($F = 5.326$, $p = 0.001$); however, the application of *C. ambrosioides* showed similar effects, with a 30% mortality rate. The application of azadirachtin and *C. subtsugae* showed 21.2% and 10% *C. tenellus* adults' mortality, respectively. At 3 DAT, the mortality was significantly greater ($F = 7.709$, $p = 0.000$) in plants treated with *Burkholderia* spp., which showed a cumulative mortality rate of 73.7%, followed by azadirachtin (61.2%). The other two treatments showed mortality rates of 56.2% (*C. ambrosioides*) and 50% (*C. subtsugae*). We recorded a similar trend at 7 DAT, where *Burkholderia* spp. produced a significantly high adult mortality rate ($F = 4.232$, $p = 0.004$) (92.5%), followed by azadirachtin (82.5%), and 80% of the cumulative mortality was due to both *C. subtsugae* and *C. ambrosioides*, which resulted in a significantly different outcome from the untreated control.

3.2. Residual Effect of Biopesticides on *Circulifer tenellus* Adults

In the residual effect experiment, no significant differences ($F = 2.085$, $p = 0.091$) were observed among treatments at 24 h post-rerelease (8 DAT). Although not significant, we recorded a higher mortality rate in plants treated with azadirachtin (25%). All products exhibited good residual activity at 10 DAT, with a cumulative mortality associated with azadirachtin of 77.5%, which was significantly greater than that of *C. subtsugae* and the control ($F = 6.922$, $p = 0.000$). *Circulifer tenellus* in the *Burkholderia* spp. treatment showed a mortality rate of 70%; in the *C. ambrosioides* treatment it was 61.2%, and mortality associated with *C. subtsugae* was 55%. All four biopesticides tested showed high residual activity at 14 DAT. However, even though no significant differences were observed among applied pesticides, they all resulted in significantly greater mortalities than the untreated control ($F = 6.126$, $p = 0.000$). In particular, the *Burkholderia* spp. and azadirachtin treatments showed a cumulative mortality rate of 93.7%, while it was 87.5% for both *C. subtsugae* and *C. ambrosioides*.

3.3. Effect of Biopesticides on *Circulifer tenellus* Nymphs

In the potency effect experiment, *C. ambrosioides*'s efficacy on *C. tenellus* nymphs was significantly higher than that of the untreated control ($F = 7.014$, $p = 0.000$), showing a mortality rate of 67.5% at 1 DAT. The other three biopesticides showed similar potencies. In particular, the mortality rates recorded were 46.2% for azadirachtin, and 42.5 and 41.2% for *Burkholderia* spp. and *C. subtsugae*, respectively. At 3 DAT, we recorded cumulative mortality rates increased in all products. Azadirachtin showed a mortality rate of 95%, which was significantly higher than *C. subtsugae* and untreated control ($F = 9.131$, $p = 0.000$). We recorded a 75% mortality with *C. subtsugae*, 81.2% for *Burkholderia* spp., and a mortality rate of 90% was associated with *C. ambrosioides*. All biopesticides tested showed high efficacy against *C. tenellus* nymphs at 7 DAT. Although no statistical differences were recorded among treatments, they led to significantly higher mortality rates than the untreated control ($F = 7.239$, $p = 0.000$).

3.4. Residual Effect of Biopesticides on *Circulifer tenellus* Nymphs

In the residual effect experiment, we did not record any statistical differences 24 h after releasing the nymphs (8 DAT; $F = 1.932$, $p = 0.113$). Residual activity at 10 DAT has been shown by high mortality rates recorded in hemp plants treated with azadirachtin (83.7%), *Burkholderia* spp. (77.5%), and *C. subtsugae* (73.7%), which were significantly higher than the untreated control ($F = 5.756$, $p = 0.000$), while the mortality rate recorded for *C. ambrosioides* was 71.2%. At 14 DAT, mortality rates showed the same trend as recorded for *C.*

tenellus adults, where it was significantly higher for all the biopesticide treatments than the control (F = 9.025, p = 0.000) but showed similar activity among the treatments. Mortality rates recorded were over 90% for all organic insecticides. In particular, we recorded a 96.2% mortality in plants treated with *C. ambrosioides*, 93.7% with *Burkholderia* spp. and azadirachtin, and 95% in plants treated with *C. subtsugae*.

3.5. Effect of Biopesticides on Tetranychus urticae

All biopesticides demonstrated high mortality rates in both potency and residual effect trials when applied to control spider mites. For the potency experiment, 24 h after treatment, we recorded a significantly high mortality (F = 17.307, p = 0.000) in plants treated with *C. ambrosioides* (81.0%), followed by *C. subtsugae* (75%), azadirachtin (65.6%), and *Burkholderia* spp. (59.4%). At 3 DAT, insecticides showed a similar trend; *C. ambrosioides* exhibited the highest mortality (88.7%), followed by *C. subtsugae* (81.9%), and *Burkholderia* spp. and azadirachtin, which affected 71.9% of the spider mites (F = 15.781, p = 0.000). At 7 DAT, there were no significant differences among the biopesticides, but all showed greater mortality compared to the untreated control (F = 2.919, p = 0.000). We recorded the following mortality rates, 92.5% for *C. ambrosioides*, 86.9% for *C. subtsugae*, 85.6% for *Burkholderia* spp., and 80% for azadirachtin.

3.6. Residual Effect of Biopesticides on Tetranychus urticae

In the residual effect experiment, at 8 DAT, *Burkholderia* spp. (71.2%), *C. ambrosioides* (63.7%), and azadirachtin (63.7%) significantly affected the mortality rate over *C. subtsugae* (39.45) and the untreated control (F = 26.606, p = 0.000). At 10 DAT all treatments showed higher mortality than the control (F = 22.803, p = 0.000). The highest cumulative mortality was associated with *Burkholderia* spp. (89.4%), followed by *C. ambrosioides* (76.2%), azadirachtin (70.6%), and *C. subtsugae* (55%). The same trend and significance in residual activity resulted at 14 DAT (F = 27.788, p = 0.000). In this case, the cumulative mortality due to *Burkholderia* spp. reached 91.0%, followed by *C. ambrosioides* (88.1%), azadirachtin (80%), and *C. subtsugae* (65.6%).

4. Discussion

All biopesticides tested in our study exhibited efficacy and residual activity against two life stages of *C. tenellus* (adults and nymphs) and *T. urticae* adults. Biopesticides based on heat-killed bacteria, such as *Burkholderia* spp. and *C. subtsugae*, are reported to have multiple modes of action and target mite and insect pests of different orders [24]. *Burkholderia* spp., a relatively new biological insecticide, showed the strongest overall response against *C. tenellus* adults in this trial, while its efficacy was moderate against nymphs at 1 and 3 DAT. To our knowledge, very little is known about the effects of *Burkholderia* spp. and the other biopesticides tested on controlling this leafhopper. *Circulifer tenellus*, which is new to hemp production, is a very well-known pest of potatoes in the U.S. Pacific Northwest. In this area, growers control *C. tenellus* using synthetic compounds prohibited on hemp, such as acetamiprid, carbaryl, clothianidin, cyfluthrin, lambda-cyhalothrin, deltamethrin, and imidacloprid, among others, whereas biological control is not a common practice despite several known natural enemies [7]. The potency effect of *Burkholderia* spp. was moderate compared to those of the other biopesticides but it showed the highest residual effect against *T. urticae*. A possible explanation could involve the effects of bacterial secondary metabolites and gut microbiome interactions. However, these mechanisms remain hypothetical and require further investigation. *Burkholheria* spp. have recently been tested against two other hemp pests. On the hemp russet mite, *Aculops cannabicola* Farkas (Acari: Eriophyidae), another common indoor hemp pest [25], *Burkholderia* spp. showed a good mortality rate at

6 DAT, and, similarly to our results, it showed a moderate residual effect at 22 DAT, where thte hemp russet mite population was halved compared to the untreated control [26]. On the corn earworm, *Helicoverpa zea* Boddie (Lepidoptera: Noctuidae), an outdoor pest that causes severe feeding damages [6], Britt and Kuhar [27] in a laboratory bioassay recorded a larvae mortality of 55% due to *Burkholderia* spp. at 4 DAT, which was significantly higher than the untreated control. The high efficacy of *Burkholderia* spp. at 3 or more DAT in adult trials for mites is not surprising as its mode of action involves exoskeleton degradation, and it takes a few days to take action. In our study, *C. subtsugae* exhibited a moderate to high potency effect against spider mites; however, its residual efficacy was lower than that of other ingredients. Golec et al. [28] evaluated the effect of *C. subtsugae*, *Burkholderia* spp., and other biopesticides on different life stages of *T. urticae*. The authors tested the mortality rates of adults at 24 and 48 h post-exposure and, in contrast to our results, they obtained a significantly higher mortality rate of *C. subtsugae* compared to *Burkholderia* spp. and azadirachtin. That being said, our results were in accordance with the mortality recorded using similar application rates in the trial with the nymphal life stage, where mortality due to *Burkholderia* spp., applied in rates of 20 and 40 mL/l, was significantly greater than *C. subtsugae* but similar to azadirachtin [28]. Contrary to information recorded for *T. urticae*, the effects of *C. subtsugae* on *C. tenellus* in this study were significantly greater than the untreated control at 7 DAT against nymphs in both potency and residual effect experiments, and at 7 DAT for the residual effect trial against *C. tenellus* adults, showing efficient potency and residual activity on this pest. *Chenopodium ambrosioides*, whose mode of action involves disruption of mobility and respiration, had the highest potency 24 h after release against *C. tenellus* nymphs as well as *T. urticae*. Musa, Međo, Marić, and Marčić [19] tested the toxicity of this *Chenopodium*-based biopesticide against different life stages of *T. urticae*, reporting no ovicidal effect but noting residual toxicity to hatched larvae, along with greater toxicity to directly treated younger immature stages compared to older ones. In field trials conducted in Virginia, *C. ambrosioides* showed a mortality rate of 40% against the brown marmorated stink bug, *Halyomorpha halys* Stål (Hemiptera: Pentatomidae), a pest recently associated with hemp [29], second only to Azera, a mix of azadirachtin and pyrethrins [30]. Moreover, *C. ambrosioides* seems to be less harmful to populations of natural enemies compared to other biopesticides [31]. Although *C. ambrosioides* and azadirachtin were more harmful to *C. tenellus* nymphs compared to adults, they were both equally effective in overall mortality of this pest in our study. Insecticides with a known deterrent effect or which reduced feeding activity, such as azadirachtin, can also help to reduce the spread of plant diseases. For example, early control of *C. tenellus*, as well as other insect vectors, would be necessary in managing transmitted diseases because a low number of vectors at the beginning of the season may reduce the risk of infective insects colonizing fields [13,23].

Hemp pests vary according to the type of production (e.g., oil, grain, or fiber), whether the plants are cultivated indoors or outdoors, and the geographic location where the plants are grown. There are limited chemical options for controlling pests on hemp, so the development of new and effective integrated pest management tactics is necessary. Due to historical controversies, hemp production and its associated pest management research have only begun to take place recently, and only a few recommendations exist for best cultivation practices or successful pest management programs [32]. Pest management in hemp is one of the most crucial research areas in the USA according to a recent national survey [33]. Biopesticides, being generally low-risk to consumers, are exempt from residue tolerances and can therefore be applied up to the time of harvest [33]. When a pest appears near harvest, a chemical may not be an option if the residue persists or is not permitted. Although some biopesticides are more costly than chemical pesticides, by limiting their use to specific situations, they can play a crucial role in reducing insecticide resistance and offer substantial long-term benefits [34].

5. Conclusions

In our study, which was conducted under greenhouse conditions, all the biopesticides evaluated exhibited significant mortality against various life stages of *C. tenellus* adults and nymphs, as well as *T. urticae* adults, thereby enhancing our understanding of pest management for these two common arthropod pests of hemp. Future research should assess the impact of these biopesticides in field-grown hemp to document their impact on arthropod pests and their natural enemies' populations. To further improve knowledge of integrated pest management programs for hemp, future work should also involve evaluation of the effectiveness of these biopesticides as rotational products under current commercial conditions.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/agrochemicals4040019/s1, Supplementary Material S1: Effects on selected biopesticides on two arthropod pests of *Cannabis sativa* L. in Northeastern Oregon; Supplementary Material S2: Caged built to release *Tetranychus urticae* onto plants.

Author Contributions: S.I.R. conceived the research; T.O. and D.I.T. conducted the experiments; T.O. analyzed the data and wrote the manuscript; S.I.R. edited the manuscript. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data collected during this study is available as Supplementary Material for this article (Supplementary Material S1).

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Schultes, R.E. *Random Thoughts and Queries on the Botany of Cannabis*; J. & A. Churchill.: London, UK, 1970.
2. Deitch, R. *Hemp: American History Revisited: The Plant with a Divided History*; Algora Publishing: New York, NY, USA, 2003; p. 244.
3. Cherney, J.H.; Small, E. Industrial hemp in North America: Production, politics and potential. *Agronomy* **2016**, *6*, 58. [CrossRef]
4. Zuk-Golaszewska, K.; Golaszewski, J. *Cannabis sativa* L.–cultivation and quality of raw material. *J. Elem.* **2018**, *23*, 971–984. [CrossRef]
5. Jeliazkov, V.D.; Noller, J.S.; Angima, S.; Rondon, S.I.; Roseberg, R.J.; Summers, S.; Jones, G.B.; Sikora, V. *What is Industrial Hemp?* Oregon State University Extension Service, EM 9240: Corvallis, OR, USA, 2019; pp. 1–6.
6. Cranshaw, W.; Schreiner, M.; Britt, K.; Kuhar, T.P.; McPartland, J.; Grant, J. Developing insect pest management systems for hemp in the United States: A work in progress. *J. Integr. Pest Manag.* **2019**, *10*, 26. [CrossRef]
7. Rondon, S.; Oppedisano, T. *Biology and Management of the Beet Leafhopper and Purple Top Phytoplasma Affecting Potatoes in the Pacific Northwest*; Oregon State University Extension Service, EM9282: Corvallis, OR, USA, 2020; pp. 1–6.
8. Bennett, C.W. *Curly Top Disease of Sugarbeet and Other Plants*; Scientific Societies: St. Paul, MN, USA, 1971.
9. Giladi, Y.; Hadad, L.; Luria, N.; Cranshaw, W.; Lachman, O.; Dombrovsky, A. First report of beet curly top virus infecting *Cannabis sativa* in Western Colorado. *Plant Dis.* **2020**, *104*, 999. [CrossRef]
10. Hu, J.; Masson, R.; Dickey, L. First report of beet curly top virus infecting industrial hemp (*Cannabis sativa*) in Arizona. *Plant Dis.* **2021**, *105*, 1233. [CrossRef]
11. Schoener, J.L. Molecular identification of plant pathogens affecting Cannabis sativa crops in Nevada. Ph.D. Dissertation, University of Nevada, Reno, NV, USA, 2021.
12. Rivedal, H.M.; Funke, C.; Frost, K. An overview of pathogens associated with biotic stresses in hemp crops in Oregon, 2019–2020. *Plant Dis.* **2021**, *106*, 1334–1340. [CrossRef]

13. Oppedisano, T.; Shrestha, G.; Anderson, S.; Thompson, I.D.; Rondon, S.I. Predicting phenology of four major hemipteran pests to enhance integrated pest management programs in potatoes in the lower Columbia Basin. *J. Econ. Entomol.* **2022**, *115*, 430–437. [CrossRef]
14. Sabelis, M.; Helle, W. *Spider Mites: Their Biology, Natural Enemies, and Control*; Elsevier: Amsterdam, The Netherlands, 1985; pp. 141–160.
15. Gerson, U.; Weintraub, P.G. Mites (Acari) as a factor in greenhouse management. *Annu. Rev. Entomol.* **2012**, *57*, 229–247. [CrossRef] [PubMed]
16. Attia, S.; Grissa, K.L.; Lognay, G.; Bitume, E.; Hance, T.; Mailleux, A.C. A review of the major biological approaches to control the worldwide pest *Tetranychus urticae* (Acari: Tetranychidae) with special reference to natural pesticides. *J. Pest Sci.* **2013**, *86*, 361–386. [CrossRef]
17. Pulkoski, M.; Burrack, H. *Twospotted Spider Mite in Industrial Hemp*; North Carolina State Extension Publications: Raleigh, NC, USA, 2020.
18. Van Leeuwen, T.; Vontas, J.; Tsagkarakou, A.; Dermauw, W.; Tirry, L. Acaricide resistance mechanisms in the two-spotted spider mite *Tetranychus urticae* and other important Acari: A review. *Insect Biochem. Mol. Biol.* **2010**, *40*, 563–572. [CrossRef]
19. Musa, A.; Međo, I.; Marić, I.; Marčić, D. Acaricidal and sublethal effects of a *Chenopodium*-based biopesticide on the two-spotted spider mite (Acari: Tetranychidae). *Exp. Appl. Acarol.* **2017**, *71*, 211–226. [CrossRef]
20. Michlig, N.; Lehotay, S.J.; Lightfield, A.R.; Beldoménico, H.; Repetti, M.R. Validation of a high-throughput method for analysis of pesticide residues in hemp and hemp products. *J. Chromatogr. A* **2021**, *1645*, 462097. [CrossRef]
21. EPA. Pesticide Products Registered for Use on Hemp. United States Environmental Protection Agency. 2022. Available online: https://19january2021snapshot.epa.gov/pesticide-registration/pesticide-products-registered-use-hemp_.html (accessed on 22 October 2025).
22. Marrone, P.G. Pesticidal natural products–status and future potential. *Pest Manag. Sci.* **2019**, *75*, 2325–2340. [CrossRef]
23. Echegaray, E.; Vinchesi, A.; Rondon, S.; Alvarez, J.; McKinley, N. Potato psyllid (Hemiptera: Triozidae) response to insecticides under controlled greenhouse conditions. *J. Econ. Entomol.* **2017**, *110*, 142–149. [CrossRef]
24. Dara, S.K. Entomopathogenic microorganisms: Modes of action and role in IPM. *eJ. Entomolology Biol.* **2017**, 1–7. Available online: https://ucanr.edu/blog/e-journal-entomology-and-biologicals/article/entomopathogenic-microorganisms-modes-action-and (accessed on 22 October 2025).
25. Cranshaw, W. Hemp Insect Pest Management: Needs and Challenges. Available online: https://webdoc.agsci.colostate.edu/bspm/InsectInformation/Talks2017/6Hemp.pdf (accessed on 22 October 2025).
26. Britt, K.E.; Kuhar, T.P. Evaluation of miticides to control hemp russet mite on indoor hemp in Virginia, 2019. *Arthropod Manag. Tests* **2020**, *45*, tsaa082. [CrossRef]
27. Britt, K.E.; Kuhar, T.P. Laboratory bioassays of biological/organic insecticides to control corn earworm on hemp in Virginia, 2019. *Arthropod Manag. Tests* **2020**, *45*, tsaa102. [CrossRef]
28. Golec, J.R.; Hoge, B.; Walgenbach, J.F. Effect of biopesticides on different *Tetranychus urticae* Koch (Acari: Tetranychidae) life stages. *Crop Prot.* **2020**, *128*, 105015. [CrossRef]
29. Britt, K.E.; Pagani, M.K.; Kuhar, T.P. First report of brown marmorated stink bug (Hemiptera: Pentatomidae) associated with *Cannabis sativa* (Rosales: Cannabaceae) in the United States. *J. Integr. Pest Manag.* **2019**, *10*, 17. [CrossRef]
30. Britt, K.E.; Kuhar, T.P. Evaluation of insecticides to control brown marmorated stink bug on hemp in Virginia, 2019. *Arthropod Manag. Tests* **2020**, *45*, tsaa055. [CrossRef]
31. Britt, K.E.; Kuhar, T.P. Evaluation of biological insecticides to manage Cannabis aphid in CBD hemp, 2019. *Arthropod Manag. Tests* **2021**, *46*, tsab057. [CrossRef]
32. Britt, K.E. Insect pest management in hemp in Virginia. Ph.D. Thesis, Virginia Polytechnic Institute, Blacksburg, VA, USA, 2021.
33. Ellison, S. Hemp (*Cannabis sativa* L.) research priorities: Opinions from United States hemp stakeholders. *GCB Bioenergy* **2021**, *13*, 562–569. [CrossRef]
34. Dara, S.K.; Peck, D.; Murray, D. Chemical and non-chemical options for managing twospotted spider mite, western tarnished plant bug and other arthropod pests in strawberries. *Insects* **2018**, *9*, 156. [CrossRef] [PubMed]